VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

22BAI1285

Final Assessment Test(FAT) - NOV/DEC 2025

Programme	B.Tech.	Semester	Fall Semester 2025-26
Course Code	BCSE418L	Faculty Name	Prof. Poonkodi
Course Title	Explainable Artificial Intelligence	Slot	E1
		Class Nbr	CH2025260102067
Time	3 hours	Max. Marks	100

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: Understand the methods and terminologies involved in Explainable AI

CO2: Differentiate the methods used in XAI and apply suitable XAI Models or approaches for given application.

CO3: Design and develop XAI use cases for real time applications.

CO4: Design of test procedures to assess the efficiency of the developed model

Section - I

Answer all Questions (7 × 10 Marks)

01. The Indian Meteorological Department (IMD) is an agency which uses an ensemble of AI models to forecast rainfall and issue warnings for Uthrahand. This prediction is based on temperature, humidity, wind speed, pressure patterns, and satellite imagery. The AI team wants to make the system explainable for different stakeholders and situations. Mr. Raju is a farmer whose wheat field is due for harvest. Ms. Kripa is a volunteer working with Disaster Management Agencies. Dr Mrutyunjay is currently the Director General of Meteorology at the India Meteorological Department(IMD). Henry is the Data Analyst working with the model in the IMD.

a) Classify the stakeholder as end users, developers and policy makers based on their roles and responsibilities. (2 marks)

b) Elaborate the type of explanations which each may be interested and justify your choice of XAI approach for each.(8 marks)

[10] (CO1/K2)

02. Given the data of a retail company that wishes to predict monthly sales in rupees (S) of its stores using a linear regression model based on amount spent on advertising in rupees (A), number of employees working in the store (E) and store area in square meters (SA). Each row corresponds to a store's monthly data, showing how independent features relate to the dependent feature.

S.No	A	E	SA	S
1	5,000	4	50	36,000
2	8,000	5	60	50,000
3	10,000	6	55	62,000
4	6,000	3	45	32,000
5	12,000	7	65	75,000

The linear regression model developed is:

$$S = 10{,}000 + 2A + 1{,}500E + 300SA$$

a) Interpret the given linear regression model in detail, explaining the impact of each coefficient in predicting the sales.(3 marks)

b) Assume the model achieves R^2 value of 65 %. How is the R^2 value calculated and what does it indicate about the performance of the model? (4 marks)

c) Identify and explain the techniques that can be used to determine feature importance in this model ? (3 marks)

[10] (CO3/K3)

03. A hospital wants to predict the risk of patient readmission within one month using patient data such as age, BMI, number of previous hospital visits, and length of stay. A Generalized Additive Model (GAM) is used to predict the outcome.

a) State the specific reason for using a GAM in this healthcare scenario rather than a standard linear or logistic regression model and describe the basic components of GAM to predict the risk of patient readmission (5 marks)

b) Explain how a GAM is interpretable and how it helps in understanding the influence of each predictor on read mission risk. (3 marks)

c) Compare and contrast GAM with Deep Learners in terms of interpretability and accuracy. Justify. (2 marks)

[10] (CO2/K3)

04. A data scientist is developing a Convolutional Neural Network (CNN) model to detect diabetic retinopathy from retinal images.

a) Identify the various levels of feature visualization that can be used to interpret the CNN model in predicting diabetic retinopathy. (3 Marks)

b) Explain how Activation Maximization is computed and how it helps in visualizing the features learned by individual neurons. (4 Marks)

c) Discuss the purpose of Sensitivity Analysis in interpreting and validating the CNN model for diabetic retinopathy detection with examples. (3 Marks)

[10] (CO4/K3)

05. Consider the following situation. It is required to lease a certain quantity of telecom bandwidth from Bangalore to Delhi. There are two service providers available 1 and 2. Provider 1 offers a direct service from Bangalore to New Delhi with a bid of 100 million rupees. Provider 1 also offers a service from Mumbai to Delhi with a bid of 30 million rupees. On the other hand, Provider 2 offers a direct service between Bangalore and Delhi with a bid of 120 million dollars and a service from Bangalore to Mumbai with a bid of 50 million rupees.

a) Set up a TU coalitional game for this situation considering the type of service and Compute the Shapley value of each player/ provider. (6 marks)

b) Plot the shapley values with a appropriate plot and discuss the impact of cost allocation among providers. (4 marks)

[10] (CO2/K3)

06. An AI powered multilingual medical assistant called "AIMediLinga" is being developed by a research team in India. The key objective of the assistant is to help patients describe symptoms in their native language. Before diagnosis the developed model must be trained to understand the semantic meaning of each word in a patient's input sentence. Interpretability of the medical assistant is assured through self-attention mechanism, identifying the relationships among tokens and appropriate weights. Each input token is associated with three key components: Query (Q), Key (K), and Value (V) vectors that determine how strongly words attend to each other. Consider the example statement: "Terrible toothache today" with the word embedding's and its parameter matrix with dimension=2. Use the below structure for softmax calculations.

Token	Word Embeddings
Terrible	[1,0]
Toothache	[0,1]
Today	[1,1]

Weight Matrix	Value
Query	[[1, 0], [0, 1]]
Key	[[1, 1], [0, 1]]
Value	[[0, 2], [1, 3]]

a) Compute the Query, Key, and Value vectors for each token. (3 marks)

b) Calculate the raw attention scores matrix between all token pairs. (3 marks)

c) Derive the normalized attention weights for each token and compute the final self-attended representations for each word. (4 marks)

[10] (CO2/K3)

07. A multispecialty hospital deploys an AI-powered diagnostic system to detect early signs of heart attack using patient ECG signals and chest X-rays. This system is trained over 10,000 patient information using deep neural network and has produced one of the given outputs: Normal, Mild Attack and Severe Attack. Ram aged 52 was

examined initially by the cardiologist and was reported to have a Normal heart function. In the subsequent review, the cardiologist was surprised to see the reports of Ram as Severe Heart Attack predicted by the AI assisted system. AI gave no visual or textual explanations for its decision and the patient is distressed about the conflicting results.

a) Explain two reasons why physicians might hesitate to fully trust AI-powered diagnostic prediction in this case, even if it has shown high accuracy during training.(3 marks)

b) Propose any two Explainable AI (XAI) methods that could help physicians interpret the AI's prediction, and describe how they reveal whether the model focused on relevant or irrelevant features.(4 marks)

c) Design a clear way to communicate the AI's reasoning to the patient to reduce anxiety while ensuring transparency. (3 marks)

[10] (CO2/K2)

Section - II

Answer all Questions (2 × 15 Marks)

08. A company G compares two explainable AI models, Model A and Model B, used to recommend skincare routines. Three evaluators rate 5 explanation aspects (clarity, justification, relevance, fairness, satisfaction) for each model on a 1–5 scale. Assume that all evaluators are trusted equally and completely.

The company uses two metrics- impression (m) and confidence (c) to measure how trustworthy and consistent the explanations are.

$$m^{P:Q} = \frac{\sum_{i=1}^{N} r_i^{P:Q}}{N}, c^{P:Q} = 1 - 2\sqrt{\frac{\sum_{i=1}^{N} (m^{P:Q} - r_i^{P:Q})^2}{N(N-1)}}$$

Where

- $r_i^{P:Q}$: rating given by evaluator P for item i of model Q
- $N = 5$(number of items/aspects)
- $m^{P:Q}$: mean impression
- $c^{P:Q}$: confidence in impression (consistency of evaluator)

Evaluator	Model A Ratings	Model B Ratings
Dermatologist (D)	4, 3, 4, 3, 4	5, 4, 5, 4, 5
Customer (C)	3, 2, 3, 4, 3	3, 3, 3, 3, 3
Brand Analyst (B)	2, 3, 2, 3, 2	3, 3, 4, 3, 4

(a) Compute the mean impression $m^{P:Q}$and confidence $c^{P:Q}$for each evaluator (D, C, B) across both models (A and B). (10 marks)

(b) Interpret the computed results. Which model is more trustworthy overall, and which evaluator shows the highest consistency? Justify your reasoning briefly based on $m^{P:Q}$and $c^{P:Q}$values. (5 marks)

[15] (CO4/K4)

09. You have trained a complex gradient boosting model to predict customer churn based on a dataset with features like last_purchase_days_ago, support_ticket_count, monthly_spend, and customer_age. Consider the instance as below.

last_purchase_days_ago = 120 days
support_ticket_count = 30
monthly_spend = ₹18,000
customer_age = 42

The GBM predicts a high churn probability of 0.82 for this customer and LIME contributions are:

Base local probability: 0.18
monthly_spend: + 0.28
last_purchase_days_ago: +0.33
support_ticket_count: +0.05
customer_age: -0.02

a) For this specific customer, explain this single customer's prediction with a LIME plot discussing the most significant features contributing to the churn prediction. (6 marks)

b) A colleague suggests that LIME's assumption of local linearity may be misleading here. Why? Explain how LIME's methodology might fail to capture the true, non-linear relationship between these features and the model's output. (4 marks)

c) Compare LIME with another local interpretability technique (SHAP) for the same case, discussing strengths, limitations, and when each should be preferred. (5 marks)